

COURSE TITLE : ROBOTICS AND AUTOMATION
COURSE CODE : 6216
COURSE CATEGORY : E
PERIODS/WEEK : 5
PERIODS/SEMESTER : 75
CREDITS : 5

TIME SCHEDULE

Module	Topics	Periods
1	Introduction of Robots	18
2	The power sources and sensors	19
3	Manipulators, actuators and grippers	18
4	Robot Programming	20
Total		75

COURSE OUTCOME

Module	G.O.	On completion of the study of this module the student will be able
1	1	To understand the industrial robots
2	1	To understand the power sources and sensors used in robots
3	1	To understand Manipulators, actuators and grippers
4	1	To understand Robot Programming

MODULE I TO UNDERSTAND THE INDUSTRIAL ROBOTS

1.1.0 To know the robot anatomy

- 1.1.1 To explain the degrees of freedom
- 1.1.2 To describe the types of manipulator joints
- 1.1.3 To explain the robot body and arm configurations
- 1.1.4 To describe the Asimov's laws of robotics
- 1.1.5 To explain Dynamic stabilization of robots.

MODULE II TO UNDERSTAND THE POWER SOURCES AND SENSORS USED IN ROBOTS

2.1.0 To list types of Power sources

- 2.1.1 To explain Power sources
- 2.1.2 To list types of sensors
- 2.1.3 To describe sensors

- 2.1.4 To explain Machine vision using Laser
- 2.1.5 To describe Machine vision using acoustic
- 2.1.6 To explain Machine vision using magnetic and fibre optic

MODULE III TO UNDERSTAND MANIPULATORS, ACTUATORS AND GRIPPERS

3.1.0 To explain Manipulators

- 3.1.1 To explain Construction of manipulators
- 3.1.2 To describe Manipulator dynamics
- 3.1.3 To describe Electronic manipulator control circuits
- 3.1.4 To explain pneumatic manipulator control circuits
- 3.1.5 To describe End effectors
- 3.1.6 To list various types of grippers

MODULE IV TO UNDERSTAND ROBOT PROGRAMMING

4.1.0 To list Types of programming

- 4.1.1 To explain Leadthrough programming
- 4.1.2 To explain Motion Interpolation
- 4.1.3 To describe Capabilities and Limitations of Leadthrough Methods
- 4.1.4 To explain Robotic languages
- 4.1.5 To explain Generations
- 4.1.6 To describe Structure
- 4.1.7 To describe Motion Commands
- 4.1.8 To explain work cell control

CONTENT DETAILS

MODULE I

Basic concepts: - Definition and origin of robotics – Different types of robotics –Various generations of robots – Degrees of freedom – Asimov's laws of robotics – Dynamic stabilization of robots.

MODULE II

Power sources and sensors : - Hydraulic, Pneumatic and Electric drives. Sensors: Sensors in Robotics- Tactile Sensors- Proximity and Range Sensors-Uses – Machine vision – Ranging – Laser – Acoustic – Magnetic and Fiber optic.

MODULE III

Manipulators, actuators and grippers : - Construction of manipulators – Manipulator dynamics and force control – Electronic and pneumatic manipulator control circuits – End effectors –various types of grippers

MODULE IV

Robot Programming: Types of programming- Leadthrough programming- - Motion Interpolation- Capabilities and Limitations of Leadthrough Methods. Robotic languages- The textual Robot Languages- Generations-Structure-Motion Commands-workcell control.

REFERENCES

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6. Mc Kerrow P.J. Introduction to Robotics, Addison Wesley, USA, 1991.
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